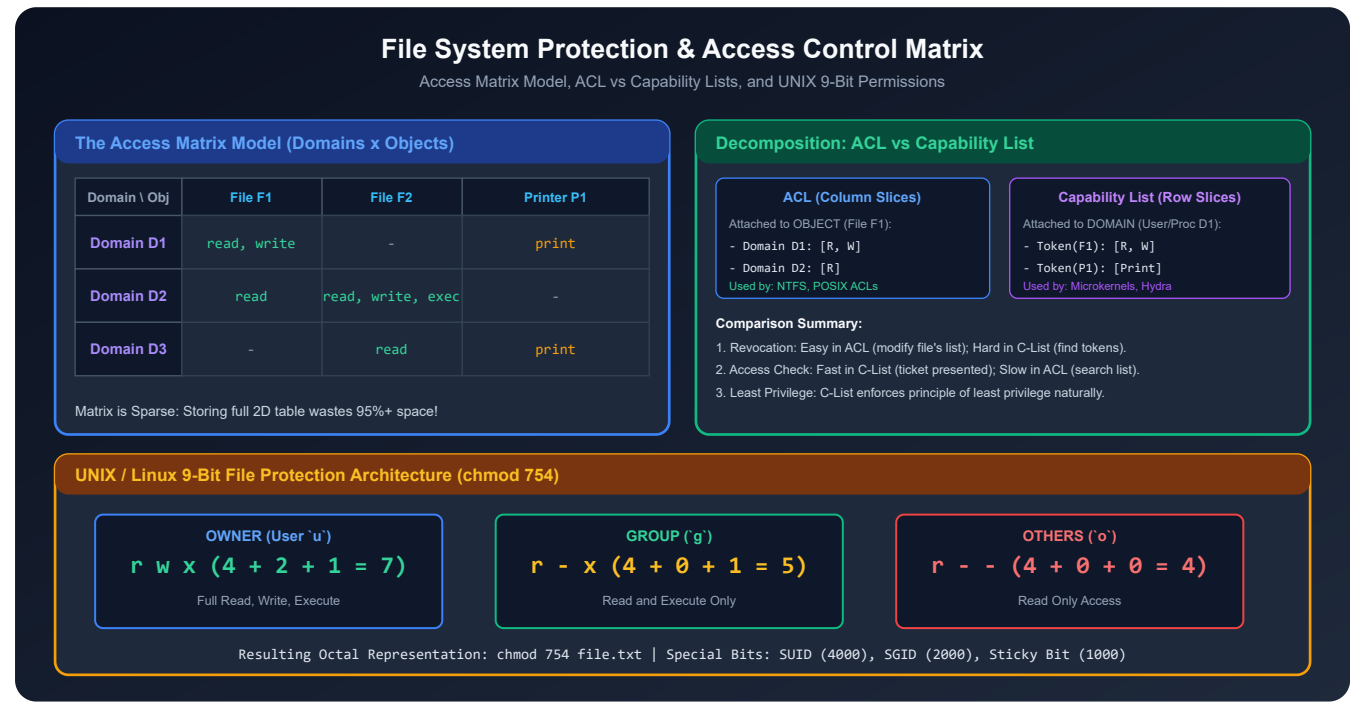


Module 13: File System Protection and Access Matrix

1. Concept & Access Matrix Model

Protection ensures unauthorized processes cannot perform prohibited operations (Read, Write, Execute, Delete) on files. The theoretical **Access Matrix** represents rights as a 2D table of Domains (Rows) × Objects (Columns).



Special Bits (12-bit mode):

- **SUID (4000):** File executes with owner's privilege (e.g. passwd binary).
- **SGID (2000):** File executes with group privileges.
- **Sticky Bit (1000):** Used in /tmp (1777); only file creator can delete their file.